TEC | Tecnológico de Costa Rica

Sistemas Fotovoltaicos

Introducción

Prof. Dr. Carlos Meza

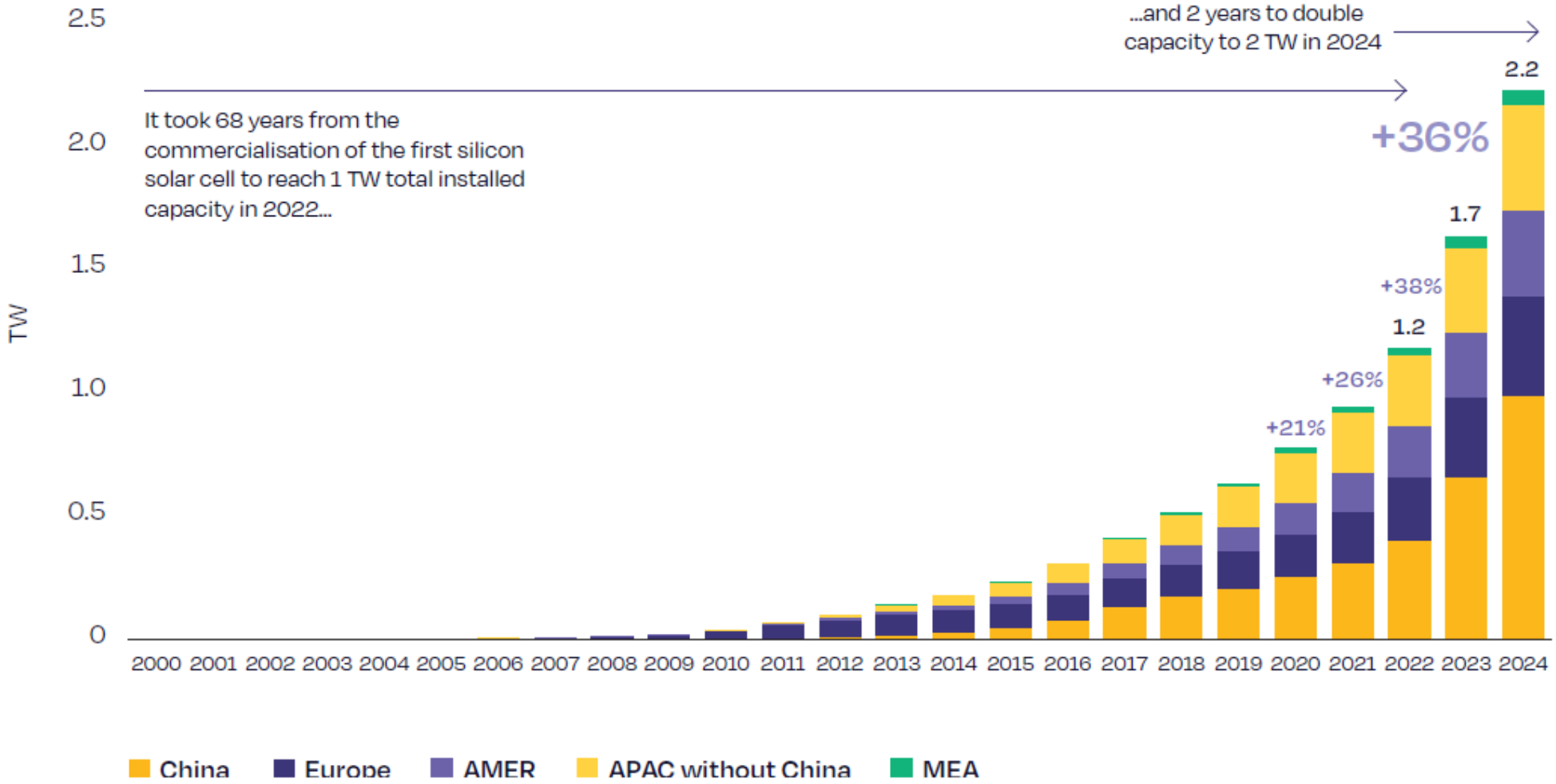
¿Por qué queremos seguir instalando plantas fotovoltaicas?

Why photovoltaic energy?

TEC | Tecnológico de Costa Rica

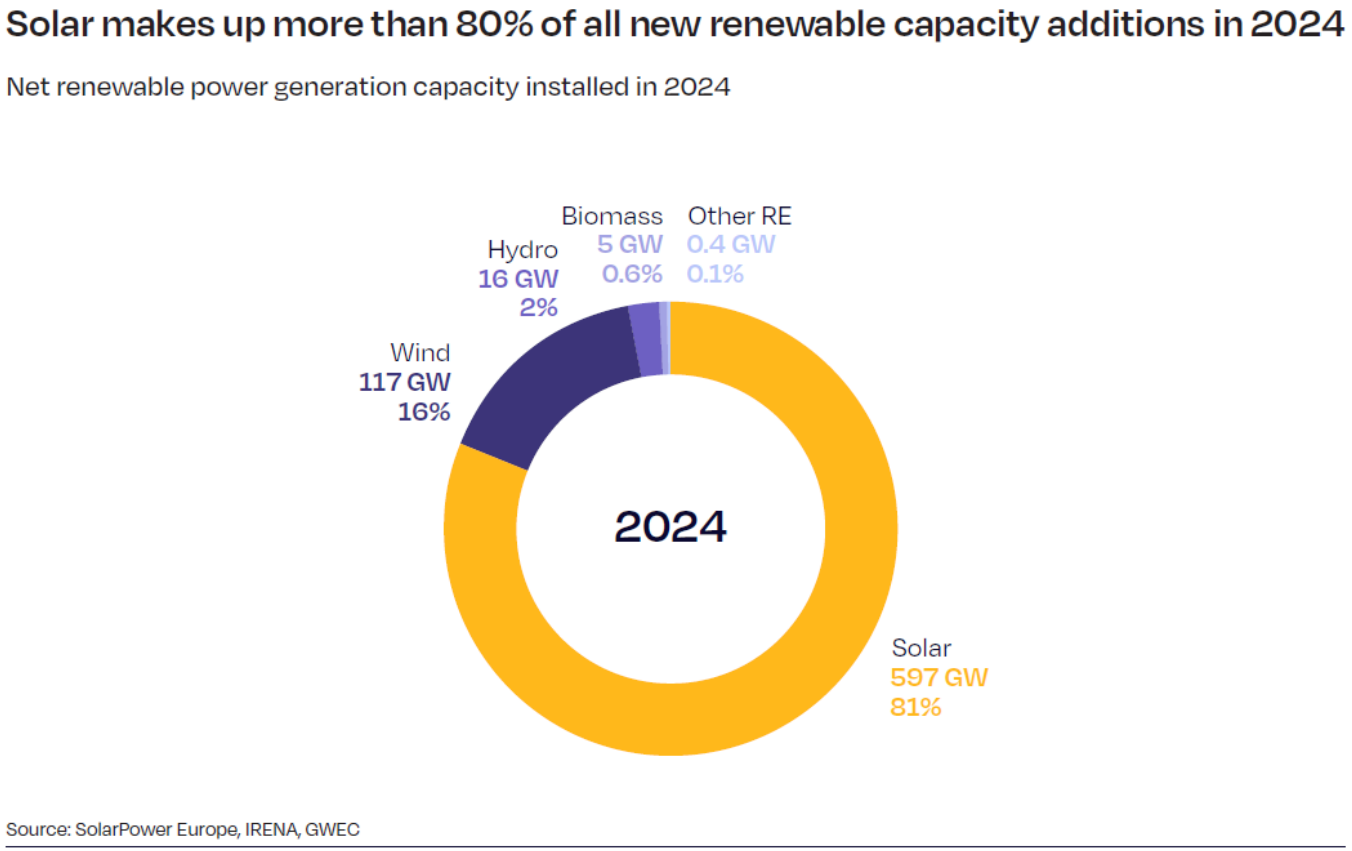
The world crosses 2 TW threshold of total solar installations in 2024

Cumulative solar PV installed capacity 2000-2024



Source: Global Market Outlook for Solar Power 2025 – 2029 (Solar Power Europe)

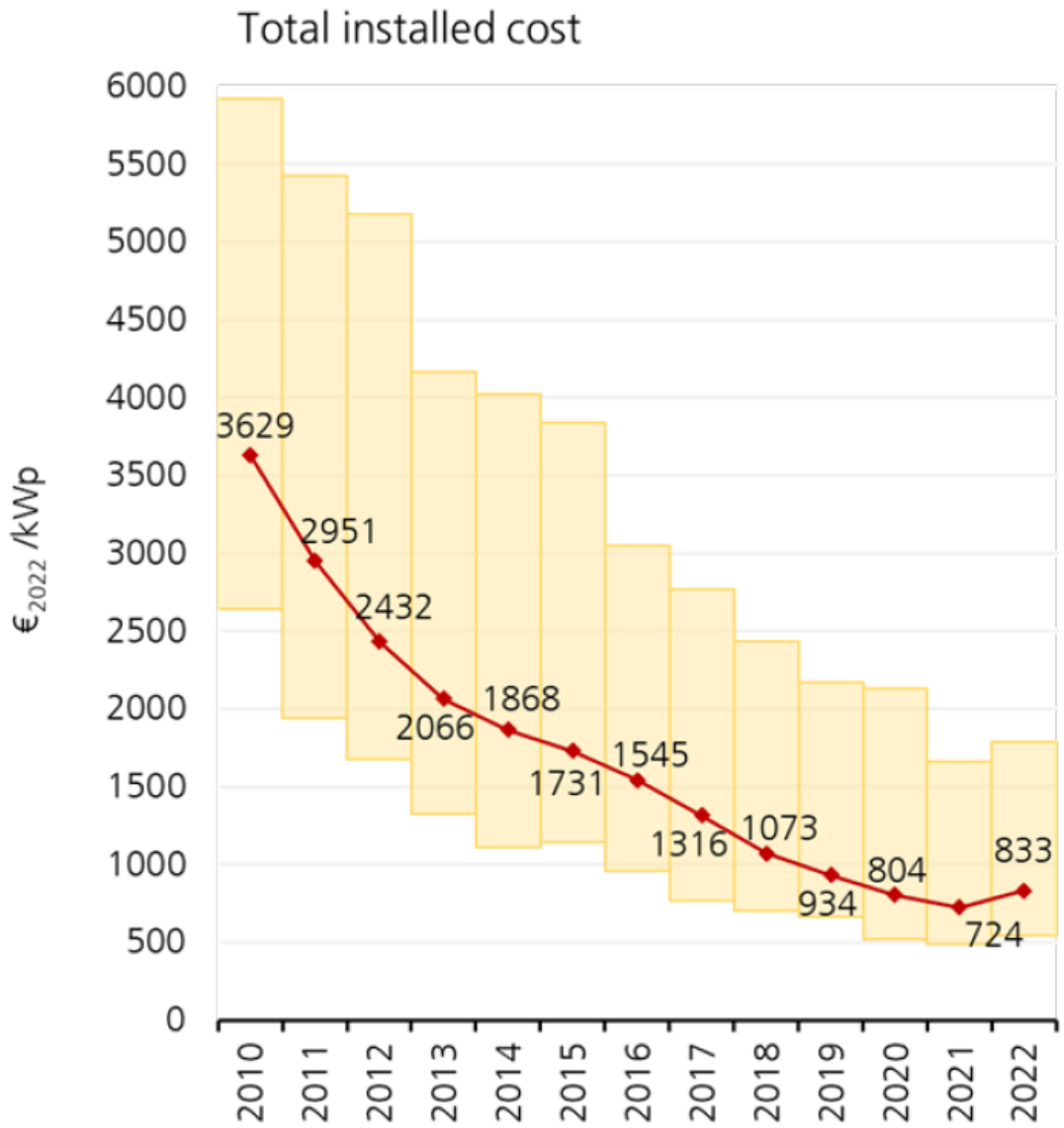
Why photovoltaic energy?



- In many countries (not all) is the cleaner and cheaper way of producing electricity.
- Available worldwide -> increase energy security.
- Versatile.
- Fastest growing energy source:
  - 2022: 300+ GWp of new solar
  - 2023: between 500+ GWp (>40% increase)
  - 2027 (estimated): 800 GWp or more.
  - According to the IEA, by 2050, photovoltaics will be the world's largest source of electricity.

Why photovoltaic energy?

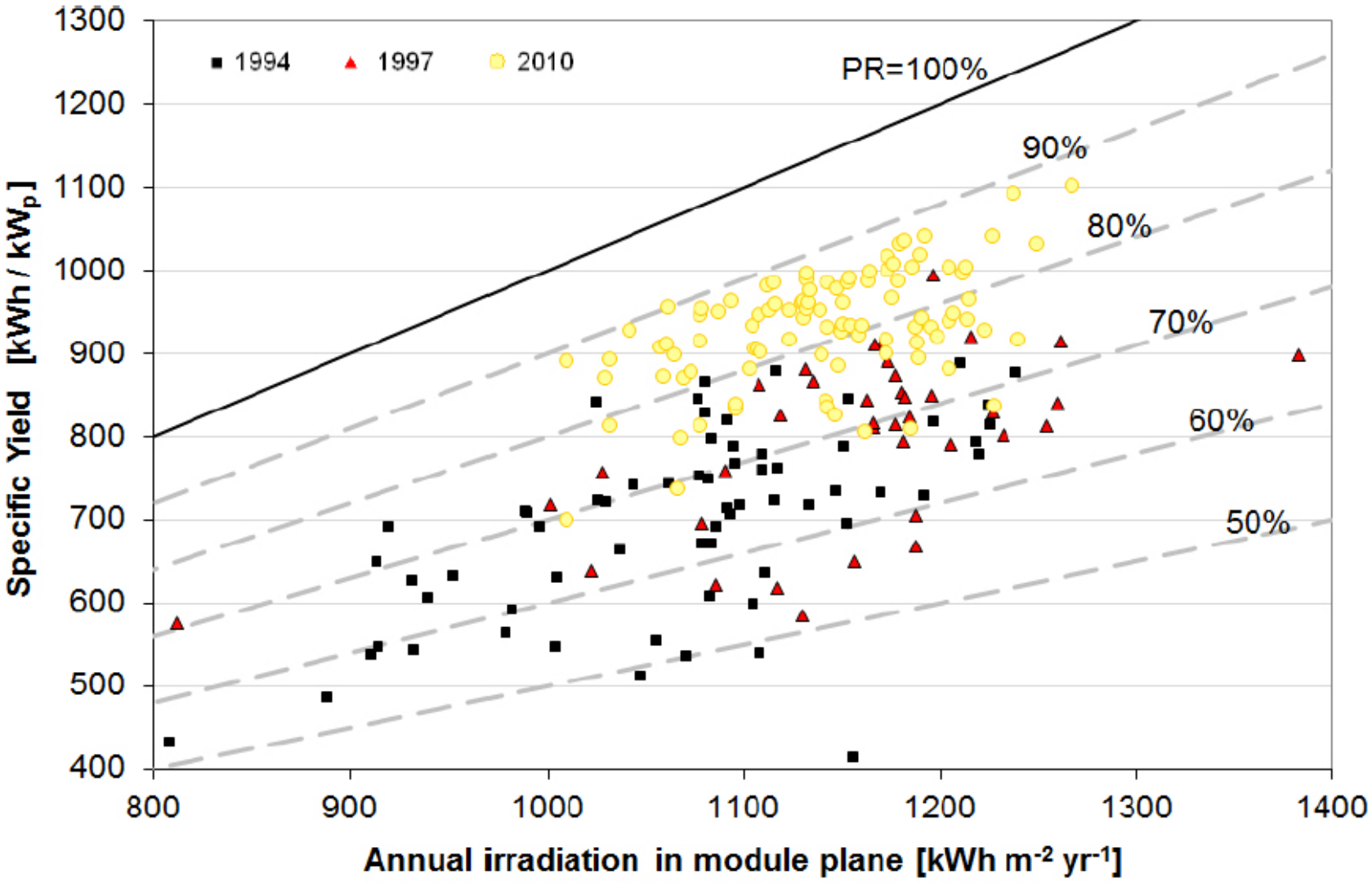
TEC | Tecnológico de Costa Rica



Data: IRENA (2023), Renewable Power Generation Costs in 2022, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: Sep-2023

The cost of large PV plants in 2022 was less than ¼ of 2010.

Why photovoltaic energy?



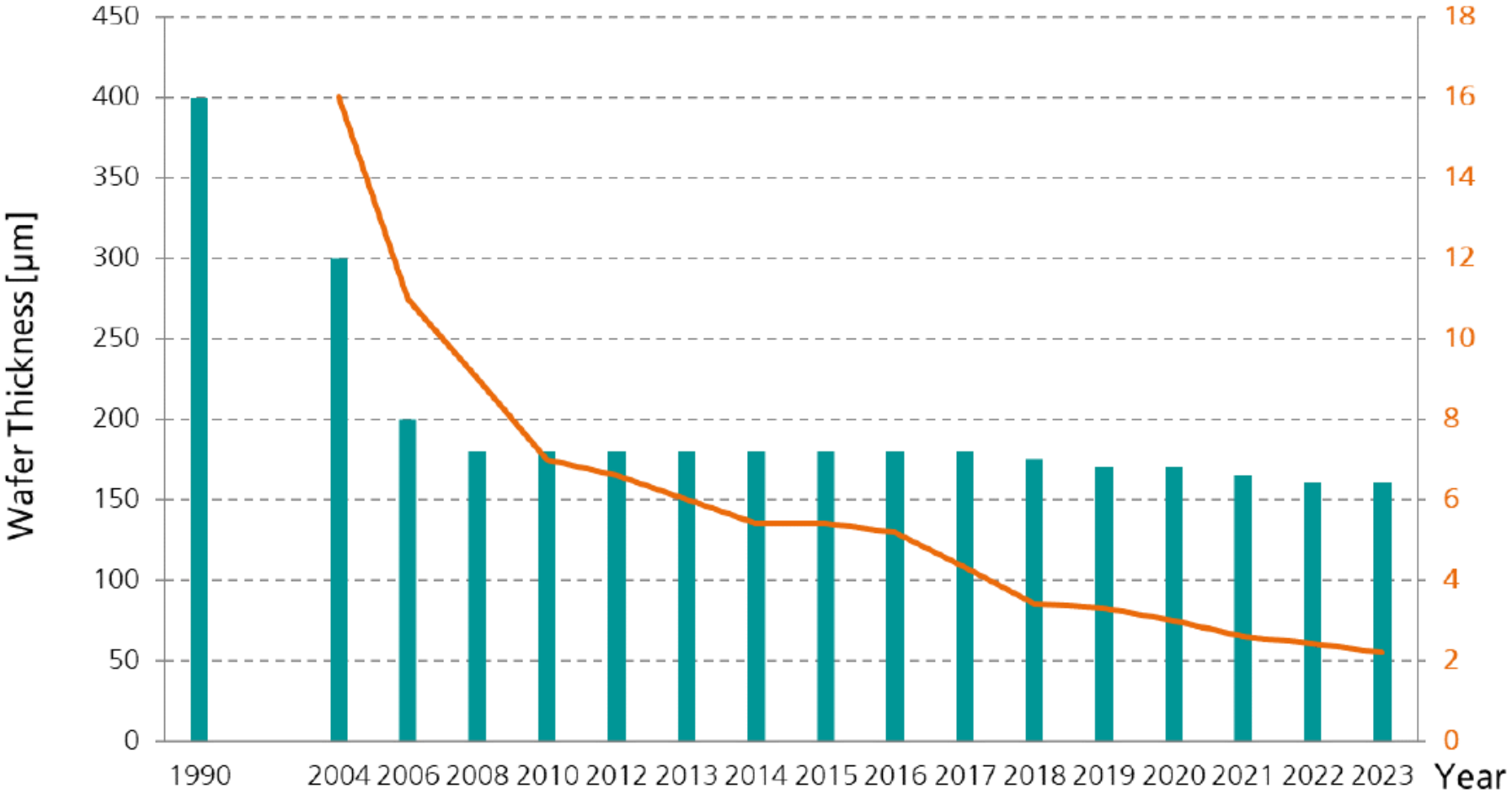
Source: Fraunhofer ISE "1000 Dächer Jahresbericht" 1994 and 1997; 2011 system evaluation, CPIA 2021

Have become more efficient

(in terms of energy production)

Why photovoltaic energy?

TEC | Tecnológico de Costa Rica

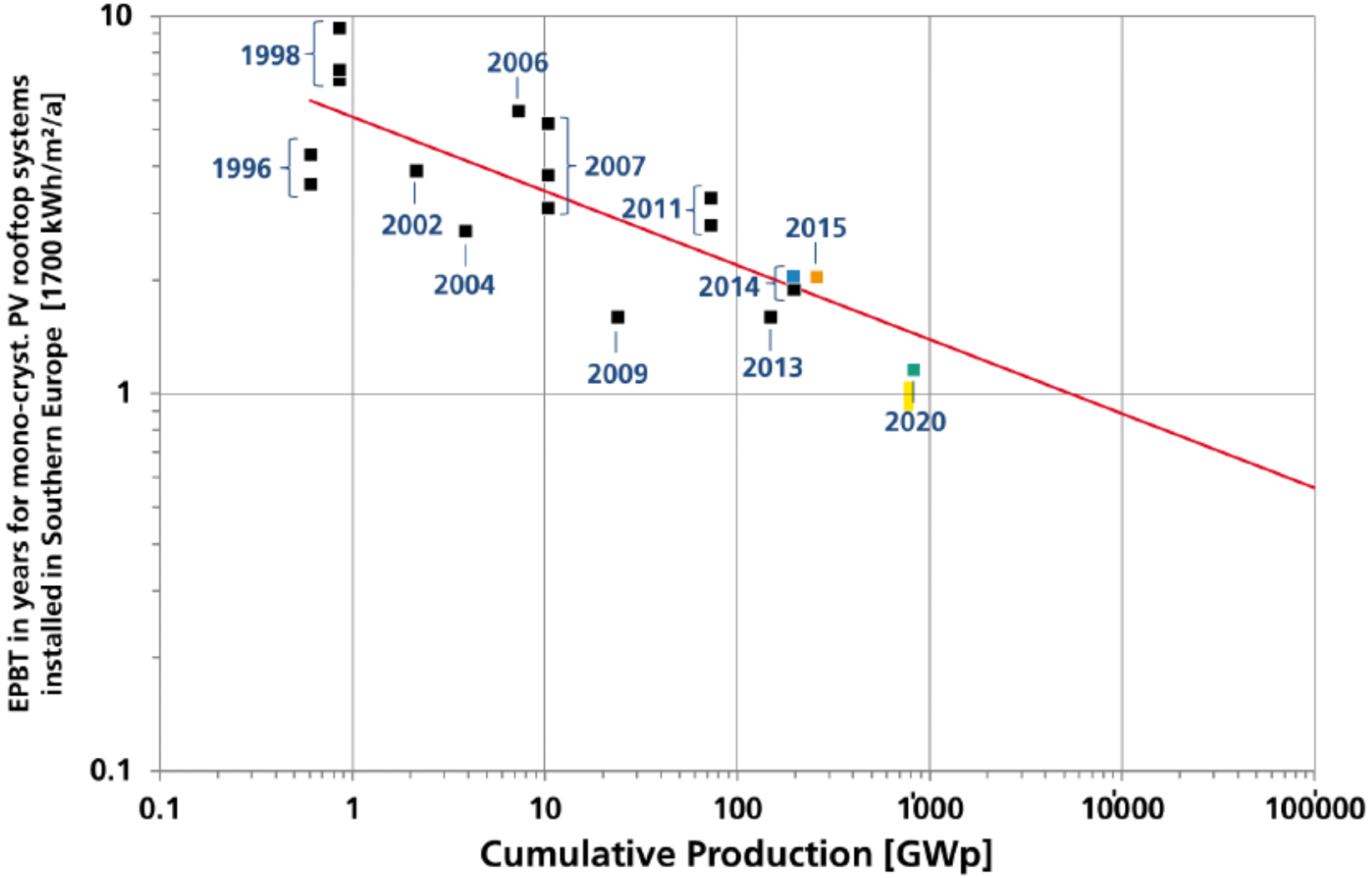


Have become more efficient

(in terms of requirements of material)

Why photovoltaic energy?

TEC | Tecnológico de Costa Rica



Irradiation: 1700 kWh/m²/a at an optimized tilt angle; **Years:** Estimated average year of original data

Have become more sustainable

Source: Fraunhofer ISE. Photovoltaics Report 2024.

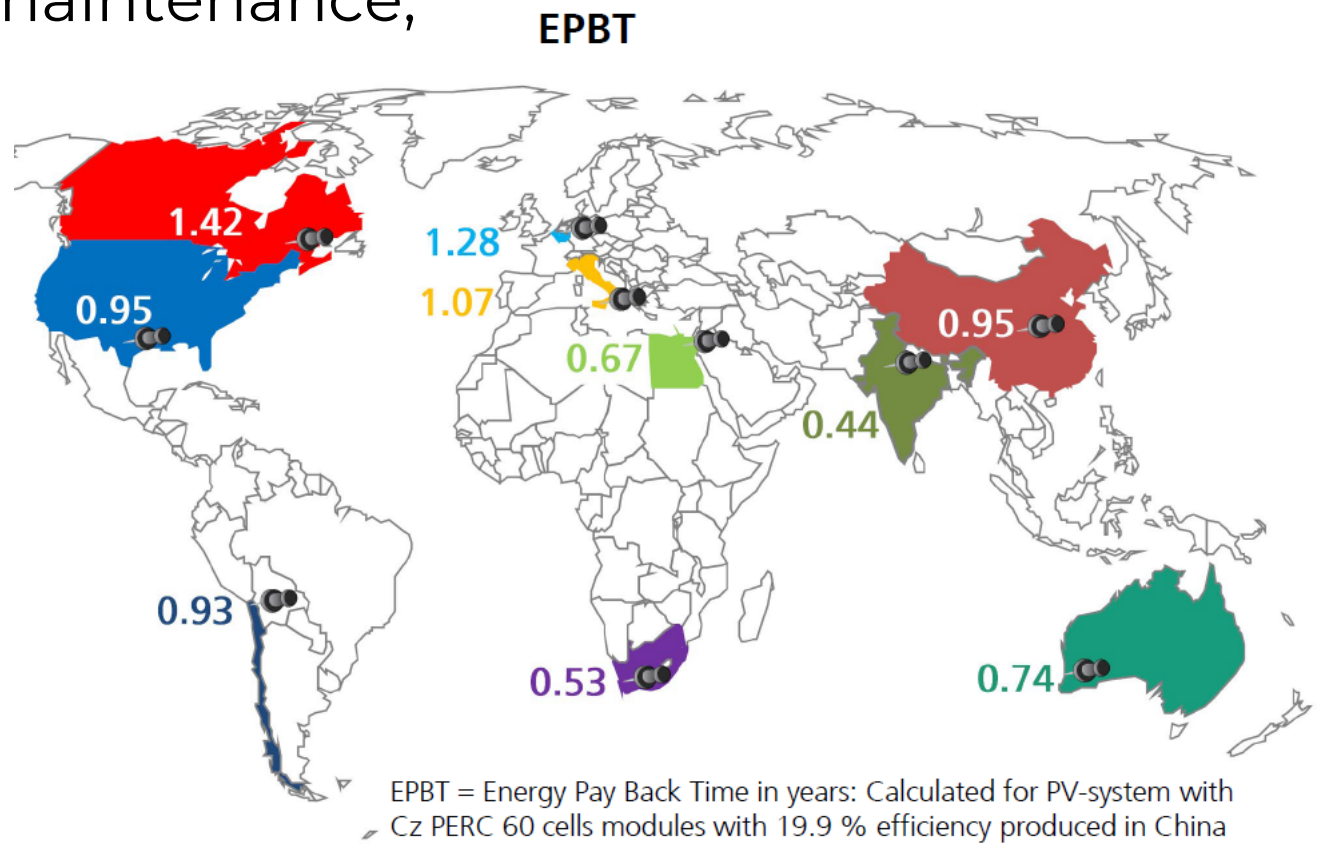
EPBT = Energy Payback Time

TEC | Tecnológico de Costa Rica

Time it takes for an energy system to generate the same amount of energy that was consumed throughout its entire lifecycle, including production, transport, installation, maintenance, and decommissioning.

Depends on irradiation, grid efficiency and other local factors

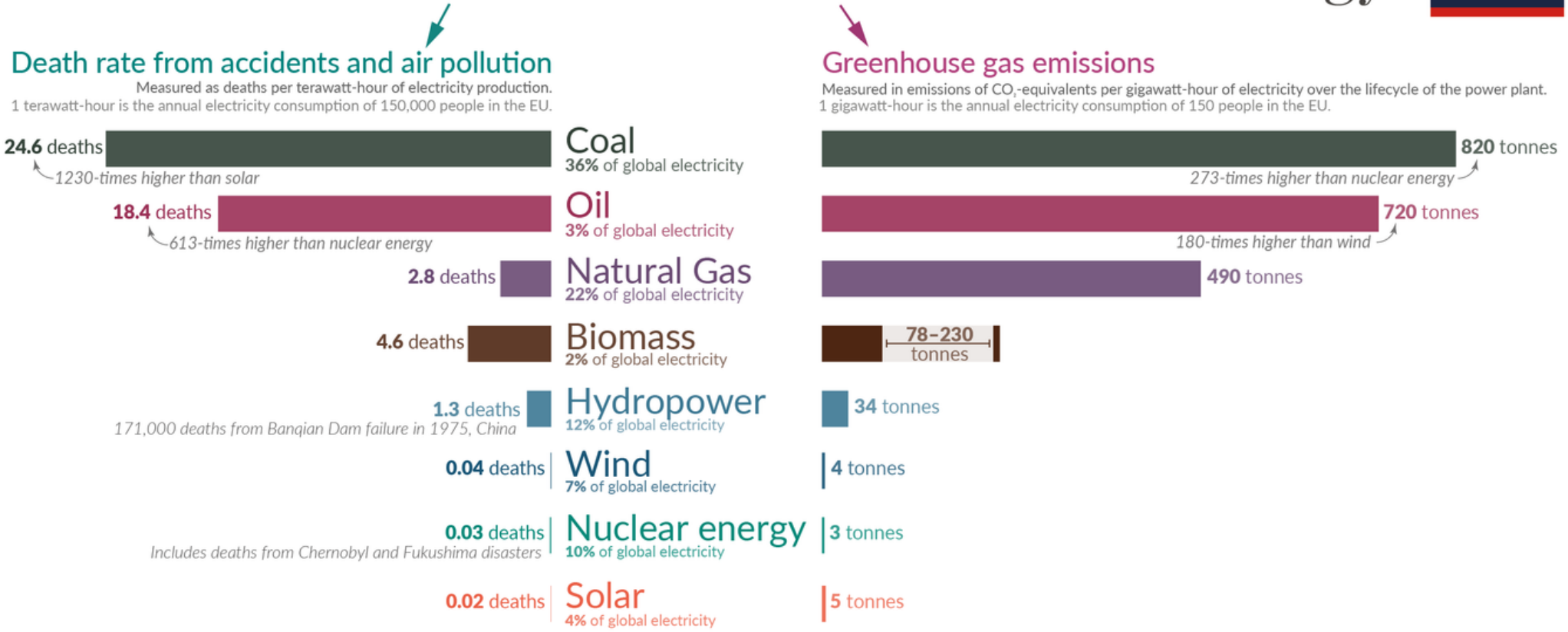
Expected lifetime of a PV system is > 20 years!!



EPBT = Energy Pay Back Time in years: Calculated for PV-system with Cz PERC 60 cells modules with 19.9 % efficiency produced in China

What are the safest and cleanest sources of energy?

Our World in Data



Death rates from fossil fuels and biomass are based on state-of-the art plants with pollution controls in Europe, and are based on older models of the impacts of air pollution on health. This means these death rates are likely to be very conservative. For further discussion, see our article: OurWorldinData.org/safest-sources-of-energy. Electricity shares are given for 2021.

Data sources: Markandya & Wilkinson (2007); UNSCEAR (2008; 2018); Sovacool et al. (2016); IPCC AR5 (2014); Pehl et al. (2017); Ember Energy (2021).

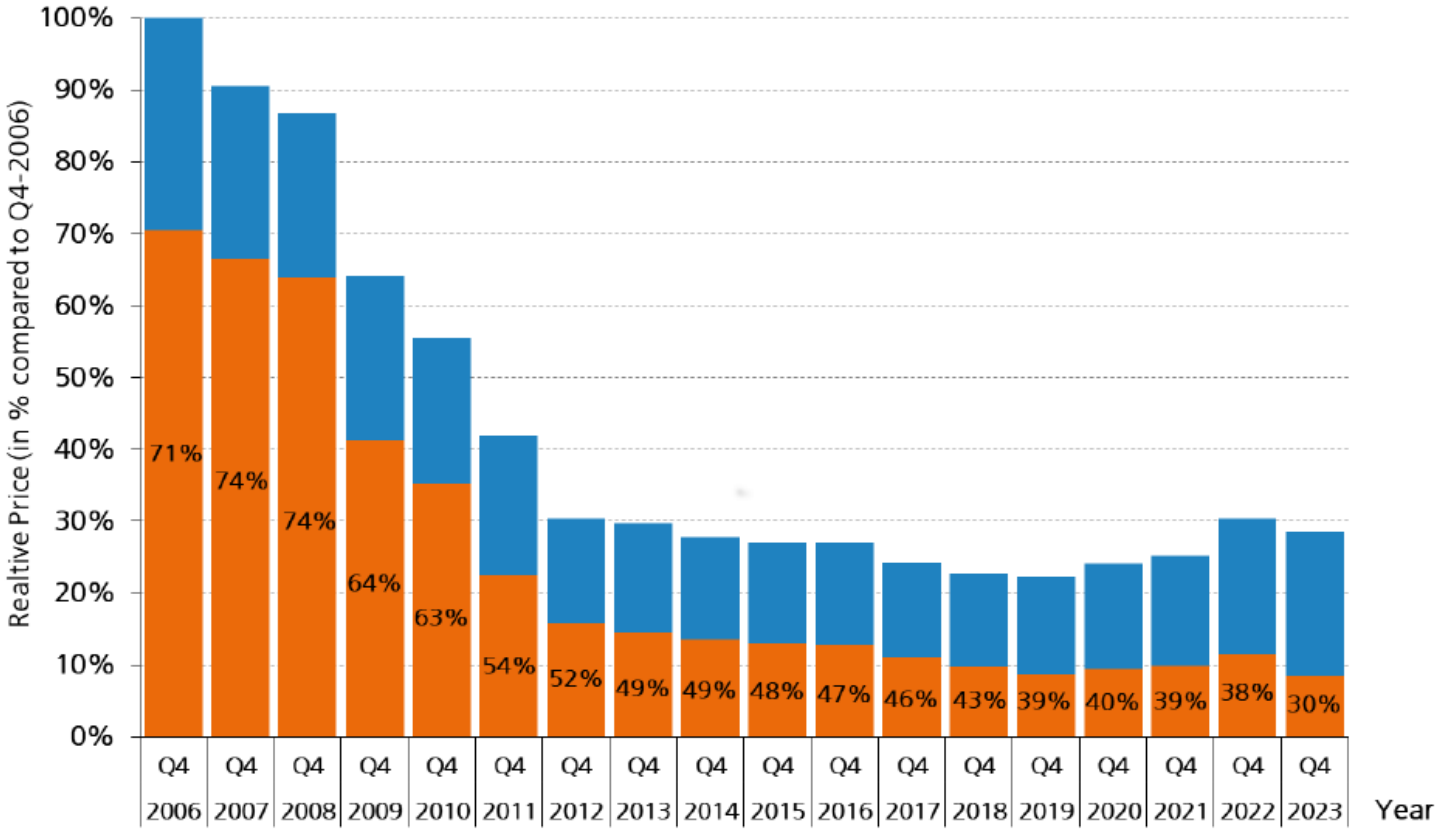
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Breakdown of cost of PV

TEC | Tecnológico de Costa Rica

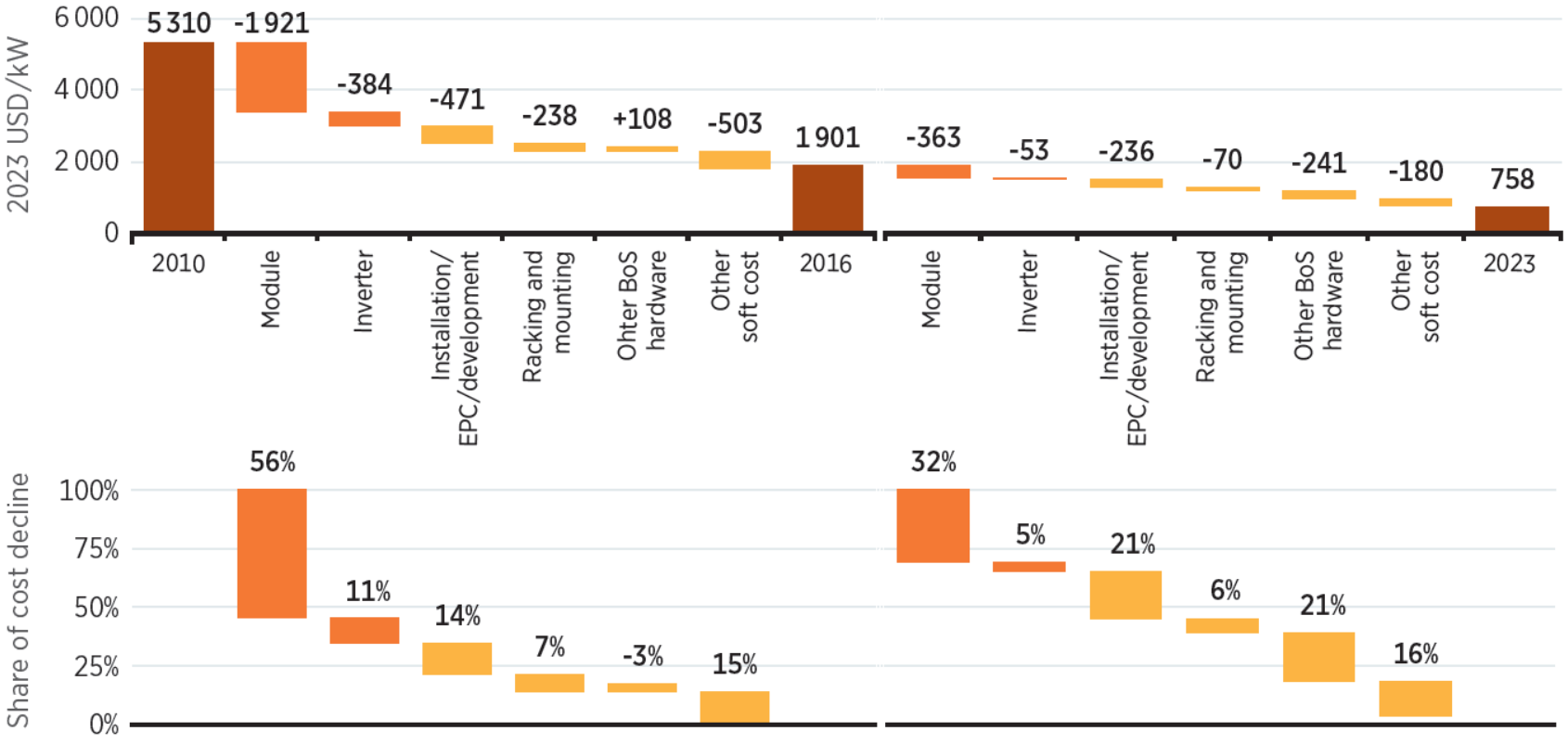
Rooftop systems (for Germany)



Balance of System (BOS) encompasses all components of a PV system excluding the PV modules. These are the inverter, mounting system, switches, wiring and installation work, for example.

Data: BSW-Solar. Graph: PSE Projects GmbH 2024. Date of data: 11/2023

Breakdown of cost of PV



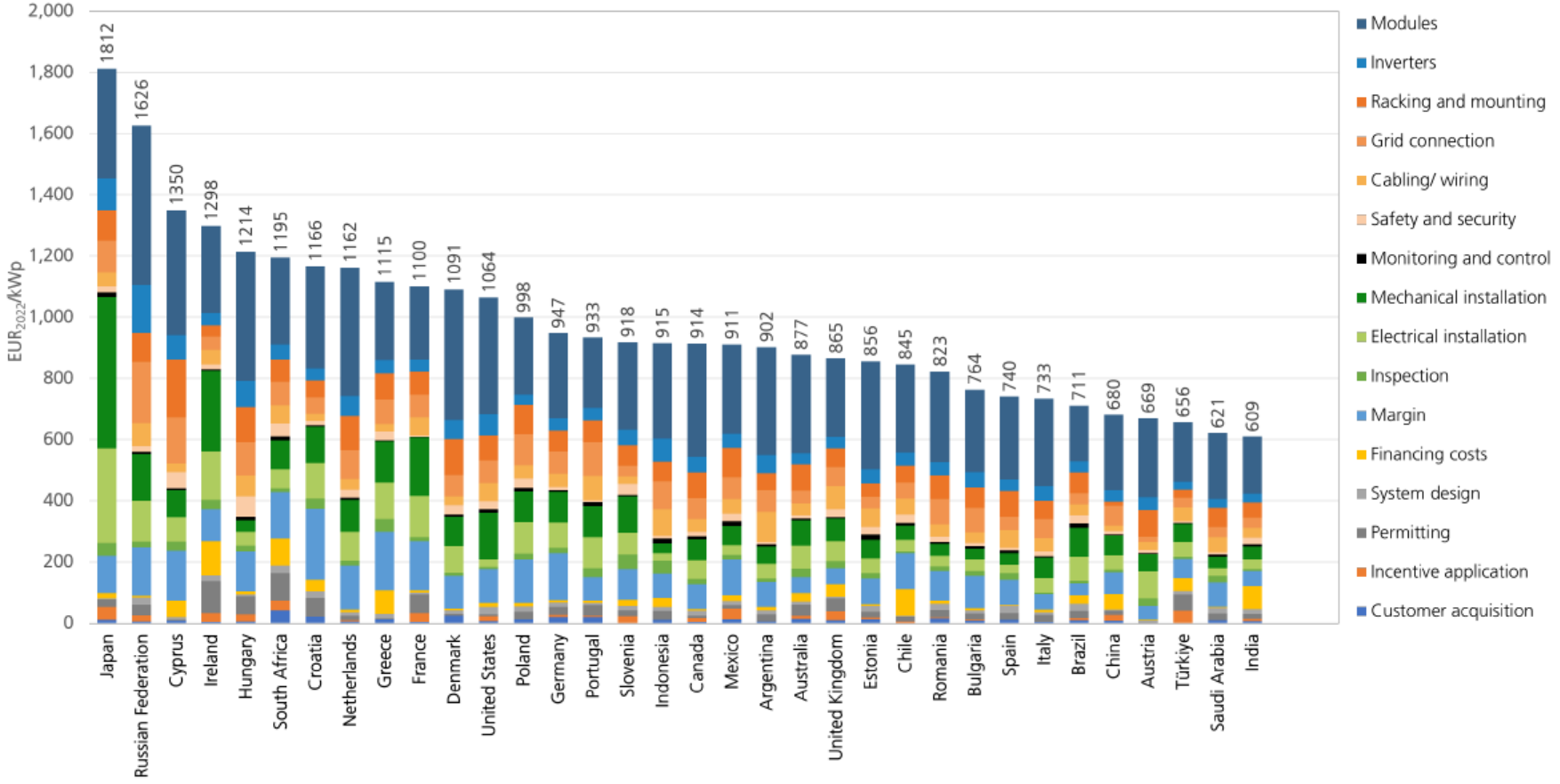
Source: IRENA

- Experiencing a continuous cost reduction.
- It is now the cheapest source of energy in many countries.

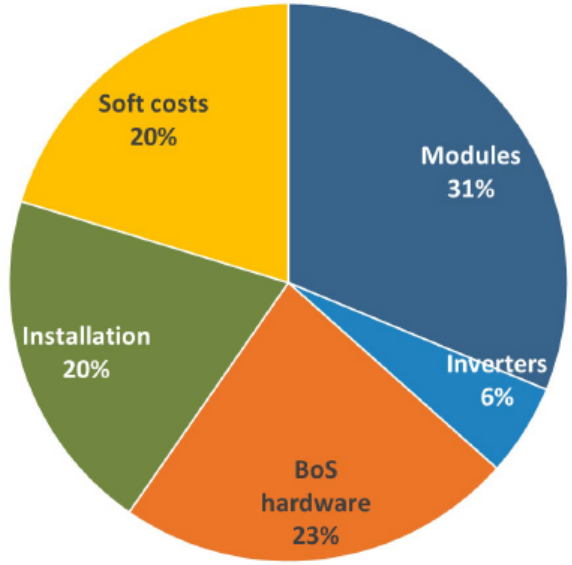
Breakdown of cost of PV

TEC | Tecnológico de Costa Rica

Utility scale (for different countries)



Breakdown of cost components
(average of available country data):



Data: IRENA (2023), Renewable Power Generation Costs in 2022, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: Sep-2023

¿De qué forma se puede sacar provecho de la energía solar fotovoltaica?

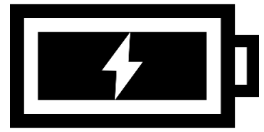
¿Cómo sacar provecho de la energía solar?



Para invertir
(vender electricidad)



Para ahorrar dinero
(reducir factura eléctrica)



Para alimentar cargas eléctricas
que no tienen acceso a la red

¿Cómo sacar provecho de la energía solar?



Para invertir
(vender electricidad)

↓

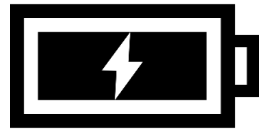
Área disponible para instalar módulos fotovoltaicos



Para ahorrar dinero
(reducir factura eléctrica)

↓

Consumo de electricidad



Para alimentar cargas eléctricas que no tienen acceso a la red

↓

Necesidades energéticas

¿Cómo y dónde se puede instalar módulos fotovoltaicos?

TEC | Tecnológico de Costa Rica

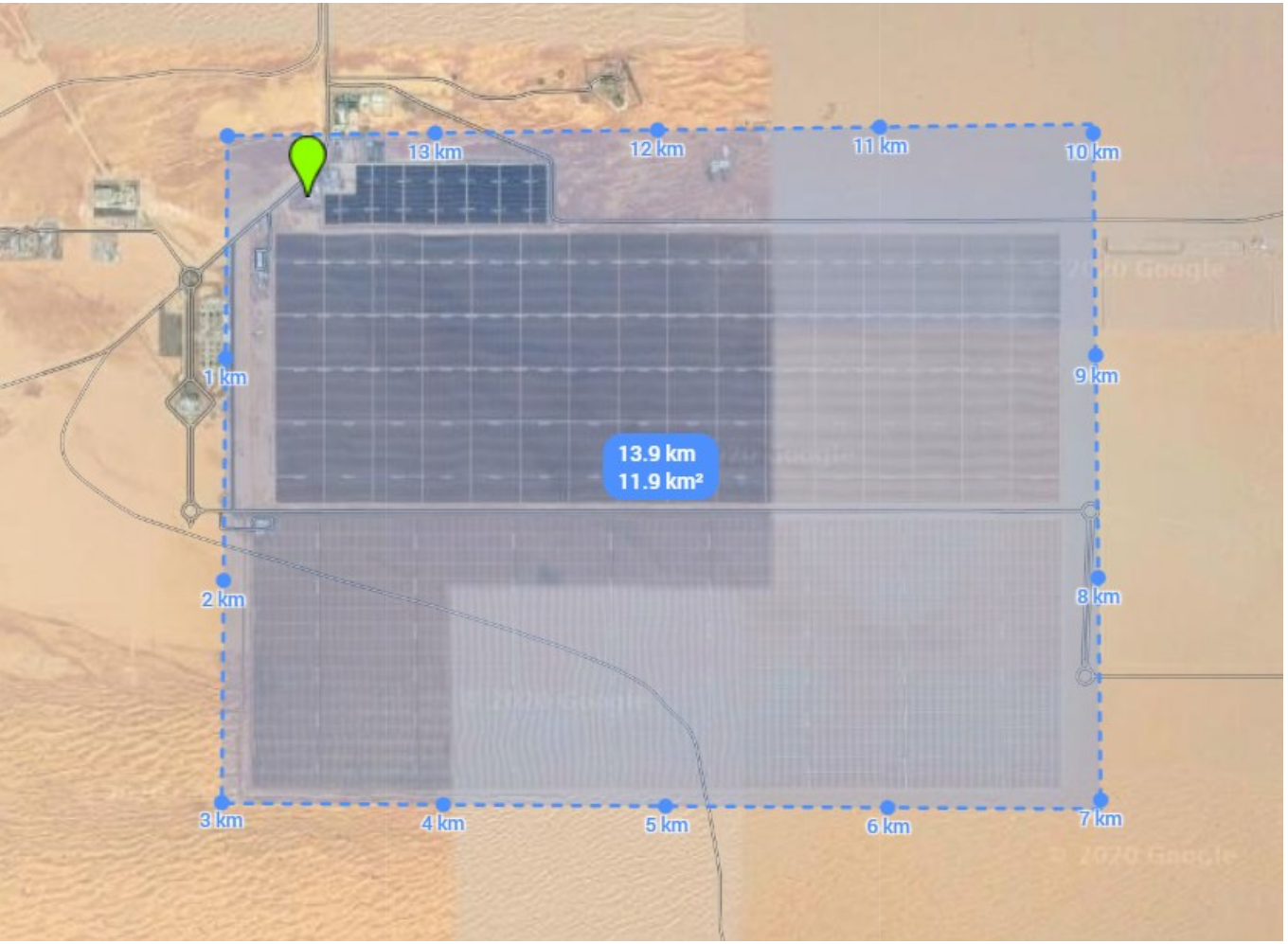
Intro & Motivation: PV systems can be BIG



Mohammed bin Rashid Al Maktoum Solar Park, Dubai, UAE

Around 4 million PV modules
Total capacity 1013 MW

Intro & Motivation: PV systems can be BIG

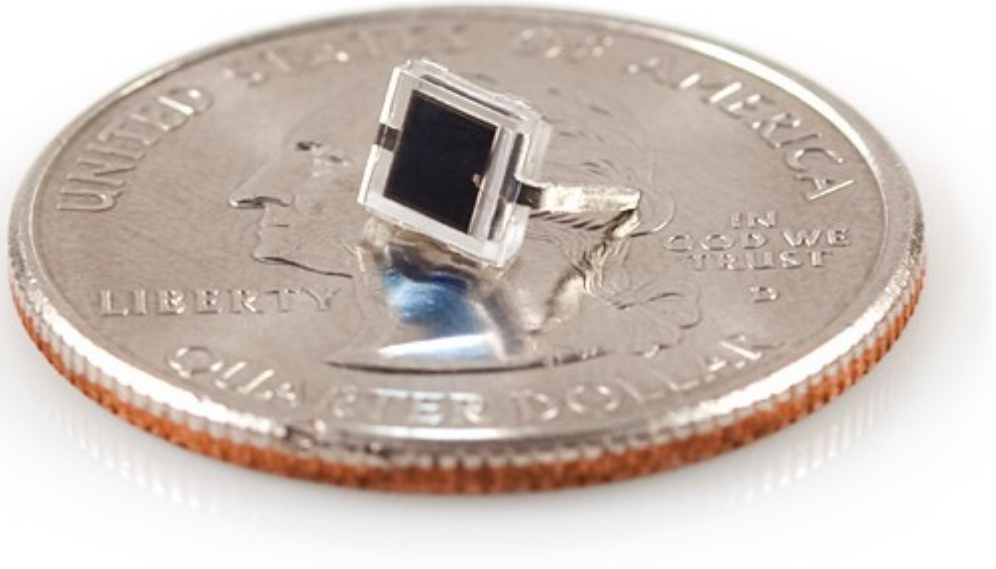


Mohammed bin Rashid Al Maktoum Solar Park, Dubai, UAE

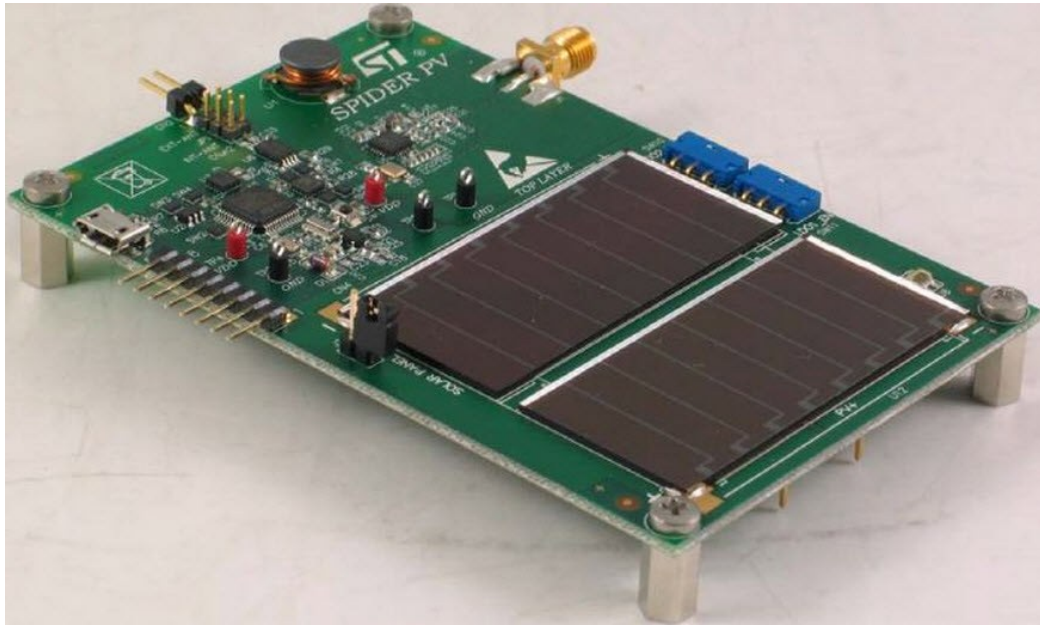
Around 4 million PV modules
Total capacity 1013 MW

TEC | Tecnológico de Costa Rica

Intro & Motivation: PV systems can be small

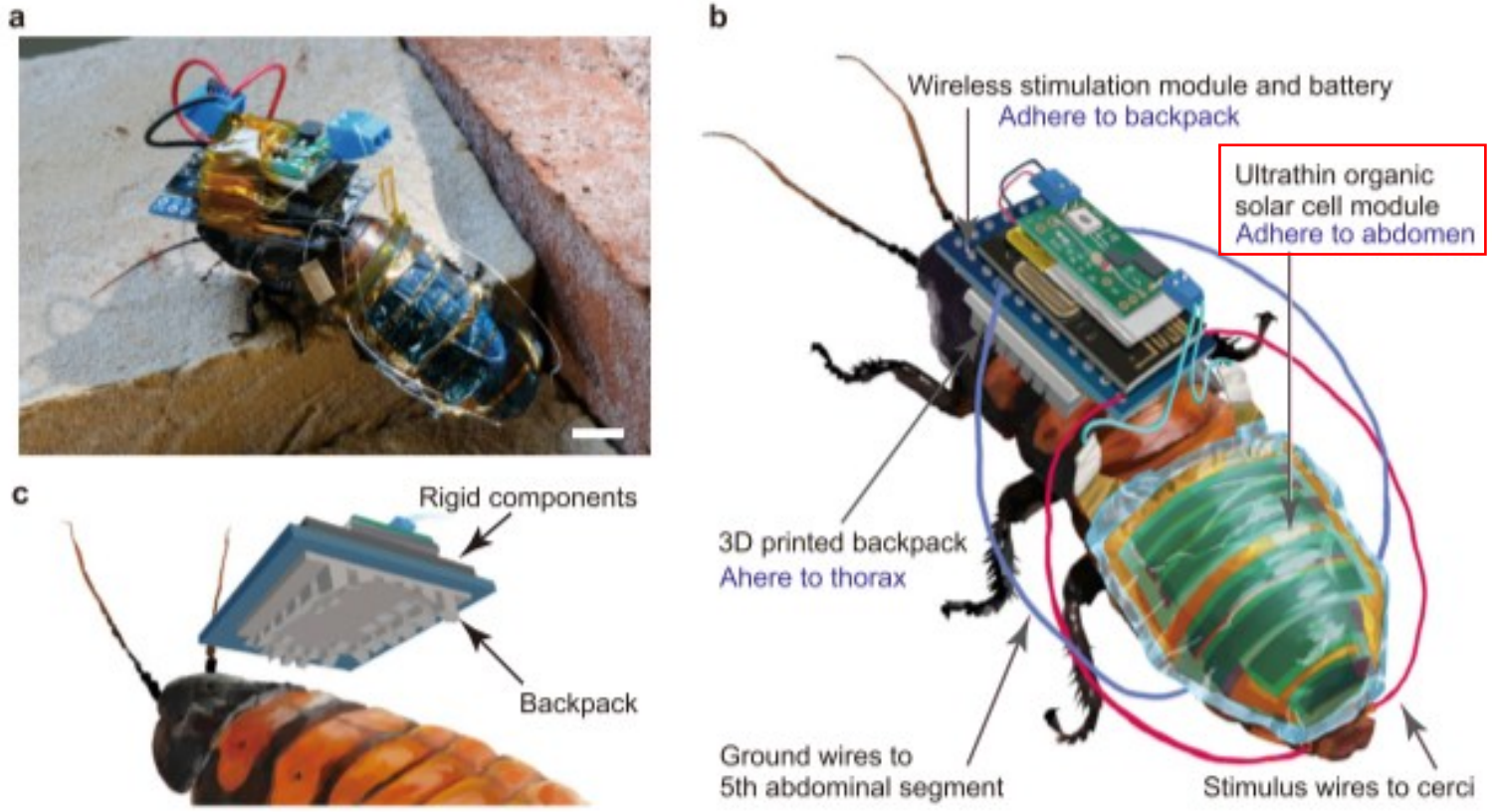


BPW34 Miniature Solar Cell



Development board for sensor nodes

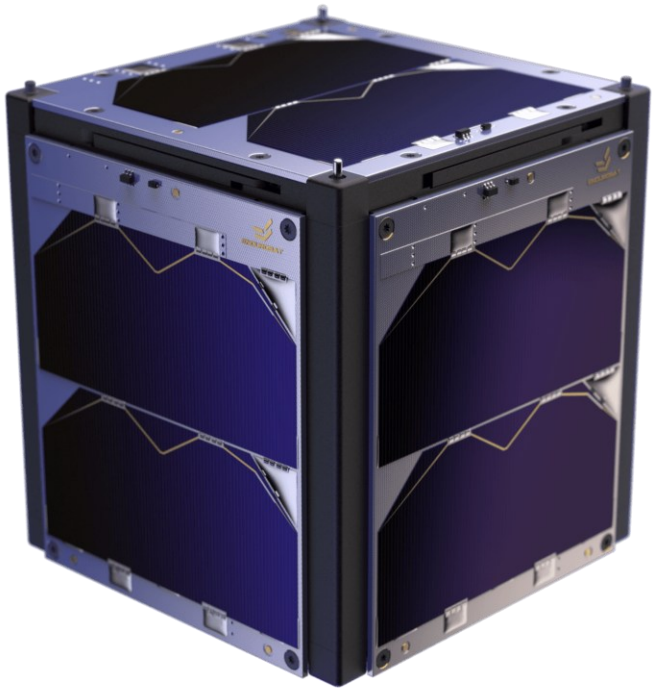
Intro & Motivation: PV systems can be small

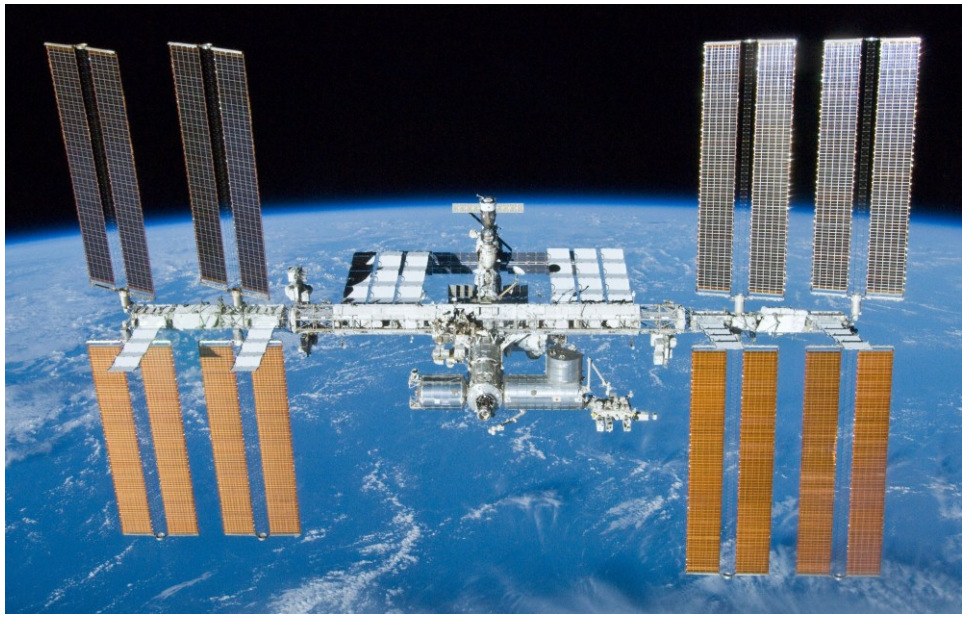


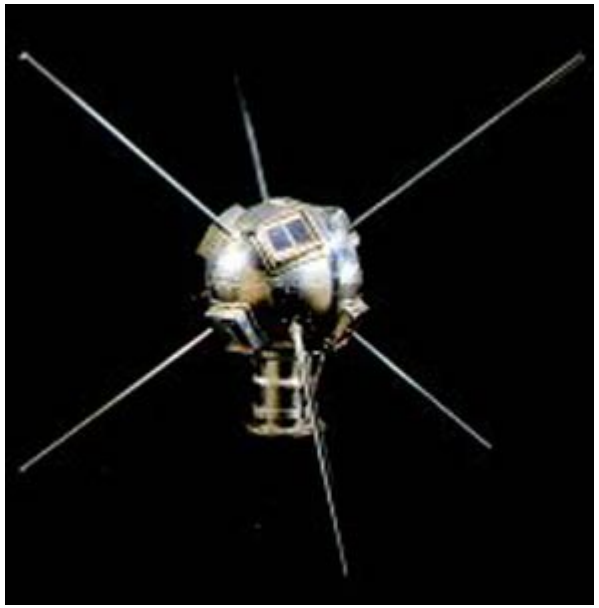
Y. Kekei et. al. Intro &Integration of body-mounted ultrasoft organic solar cell on cyborg insects with intact mobility Motivation: PV systems can be small. Flexible Electronics, 2022.

TEC | Tecnológico de Costa Rica

Intro & Motivation: Space applications



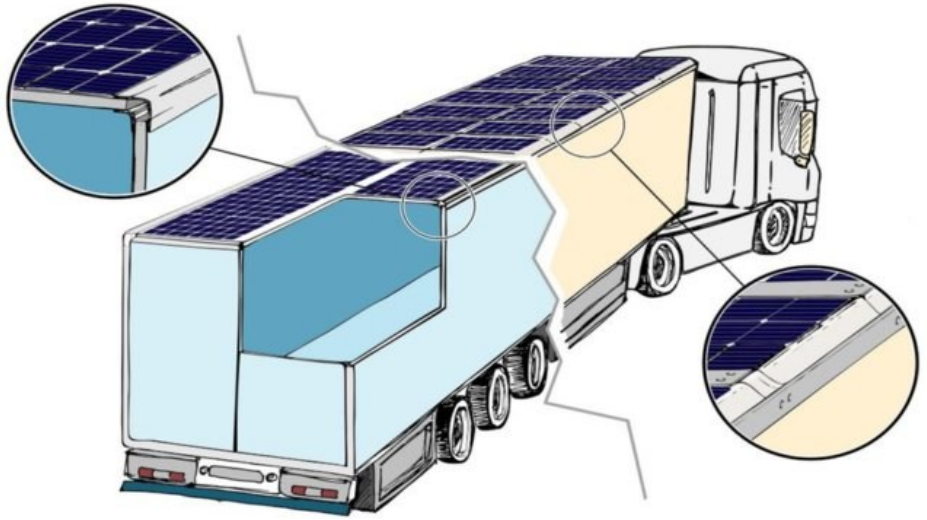




Intro & Motivation: Vehicle integrated PV



Several commercially available family cars already offer PV-integrated panoramic sunroofs, either standard or on their option list. From left to right: 2019 Prius Prime (Image ©Toyota), 2020 Revero (Image ©Karma Fischer), 2021 Ioniq 5 (Image ©Hyundai) – Source: https://www.energyville.be/



VIPV Panel Laminate of Fraunhofer (Fraunhofer ISE 2019)

Intro & Motivation: Vehicle integrated PV



Sunseeker DUO (Solar Flight)

Intro & Motivation: Vehicle integrated PV



Silent 55 (Silent Yacht)

Intro & Motivation: Agrivoltaics



APV system installed in Heggelbach, Germany



Vertical bi-facial solar modules from the company- Next2Sun GmbH

Intro & Motivation: Robot integrated PV






Tertill solar-powered weeding robot, robot “RIPPA” on a vegetable farm in Lindenow, solar-powered VitiRover Agricultural drone for pesticide spraying.

Intro & Motivation: Floating PV



Die größten schwimmenden Solarparks der Welt erreichen Leistungen von mehr als 100 MW Höchstleistung und befinden sich in China. Einer von ihnen ist seit November 2017 auf einem See in Guqiao in der Provinz Anhui im Osten des Landes installiert.
Source: https://www.en-former.com/

Intro & Motivation: Road integrated PV

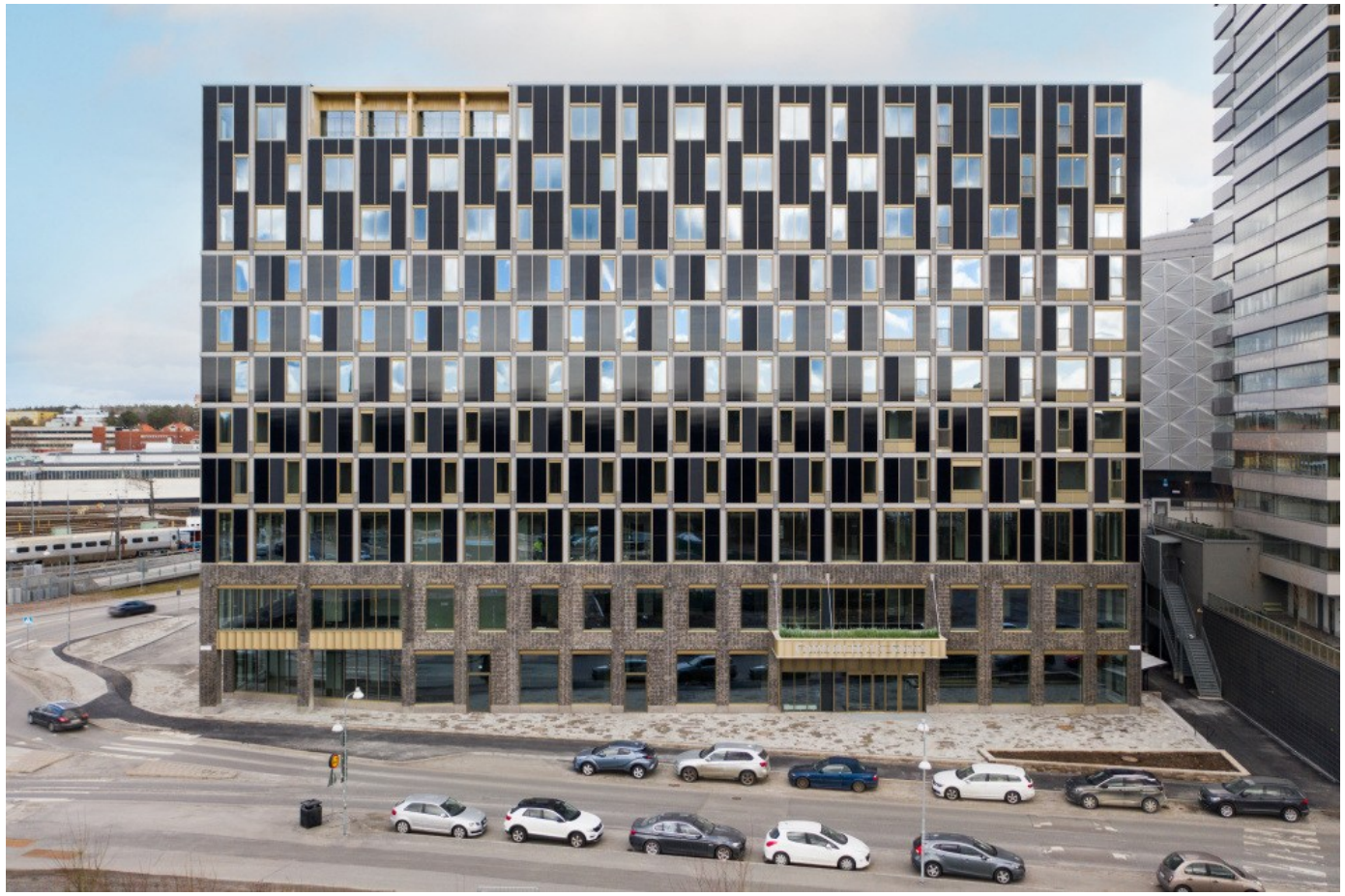


Fraunhofer Institute for Solar Energy Systems ISE, and the Austrian Institute of Technology



Noise barriers - Mitrex

Intro & Motivation: Building integrated PV



712 glass-glass BIPV modules of ML System are installed at Aeres Hogeschool Almere (Netherlands)

Intro & Motivation: Urban integrated PV



From SpotlightSolar ®

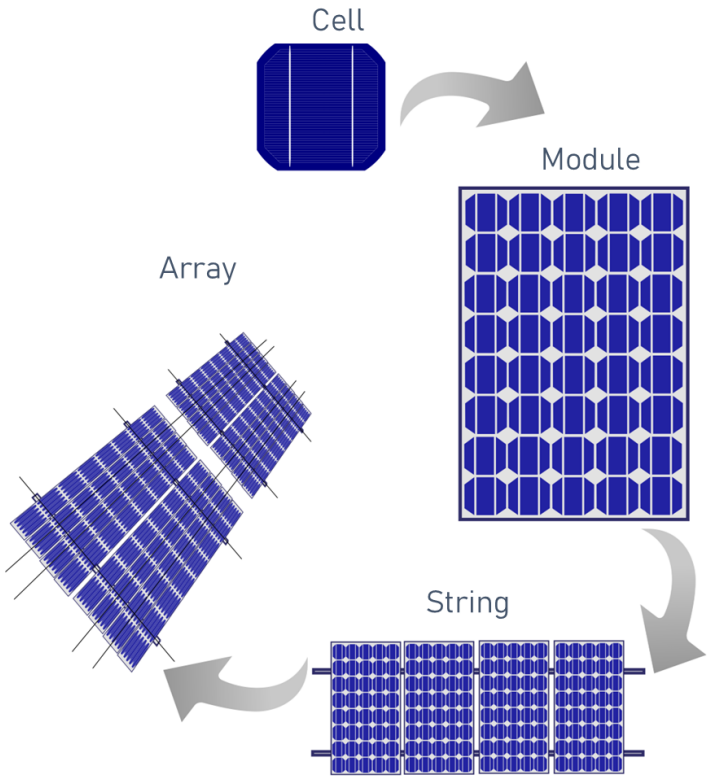
Intro & Motivation: Urban integrated PV

... and many more



From SpotlightSolar ®

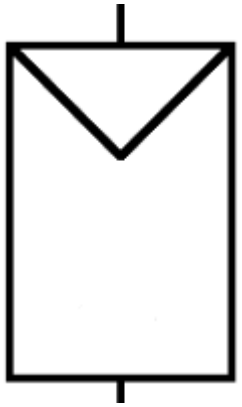
¿Cuáles son los principals components de un systema fotovoltaico?



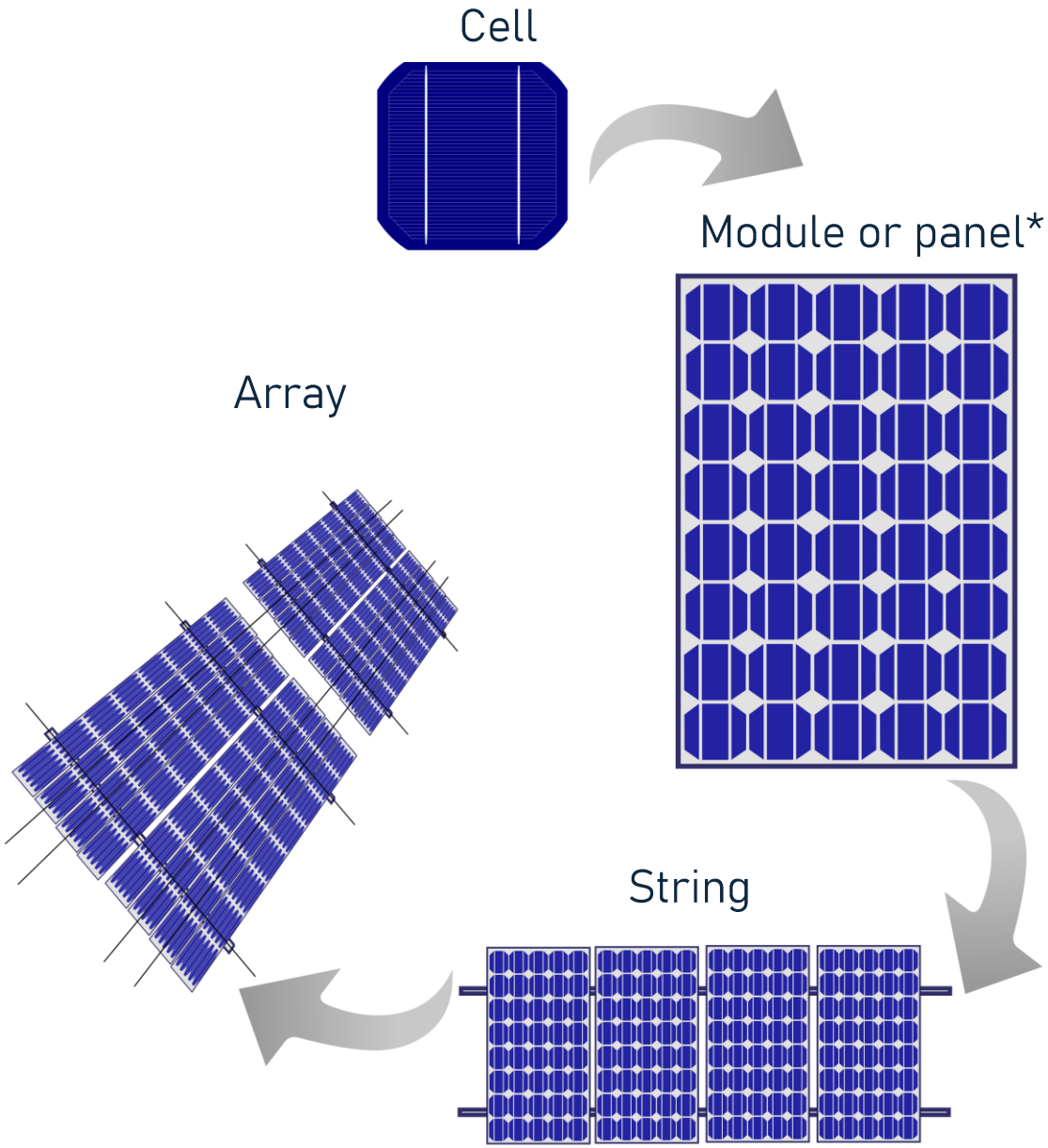


(Diagram by Carlos Meza modified from an image from Rfassbind under license CC BY-SA 4.0)

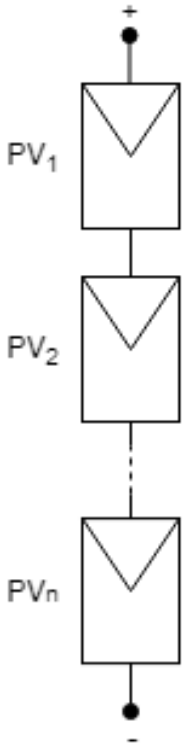
PV Arrays and Strings



Schematic symbol for a PV module



PV String

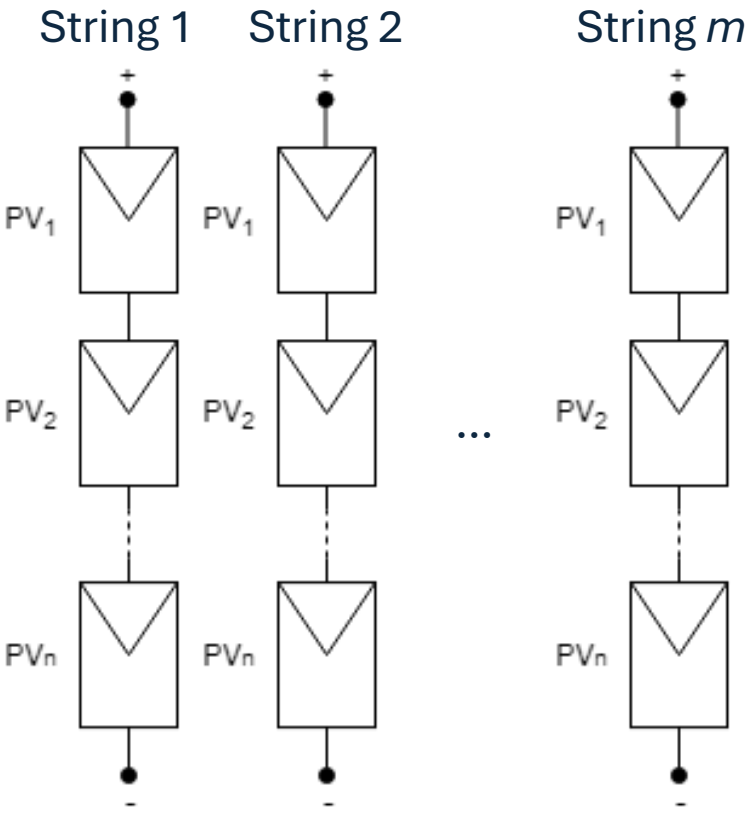


When all the panels are identical, they have the same irradiance and temperature:

For n panels connected in series (string):

- They will have a voltage equal to n times that of a panel
- The same current will flow through them
- Power equal to n times the power of a panel

PV Array



Array of *m* strings.
Each string has n PV modules

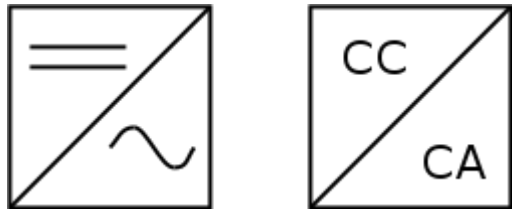
TEC | Tecnológico de Costa Rica

Photovoltaic Inverters

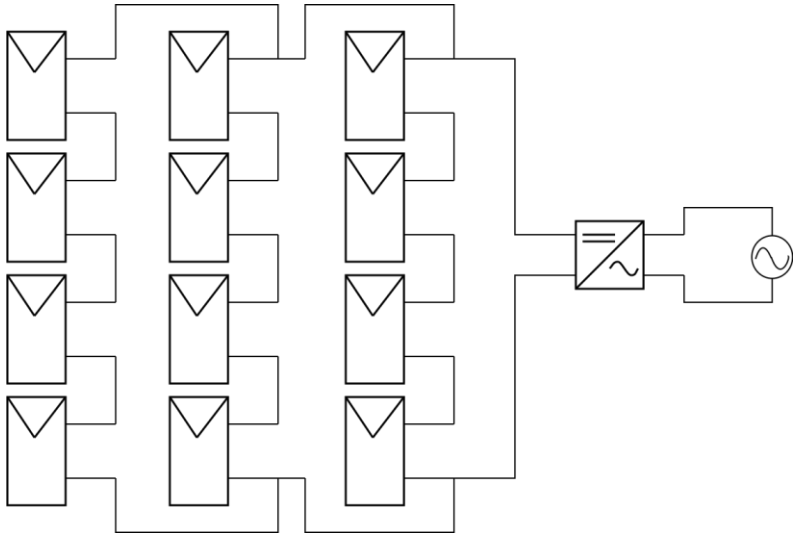
PV Inverter

Tecnológico de Costa Rica

Symbols:



Example of a simple schematic:





COAL Power plant

Hydro Power **plant**

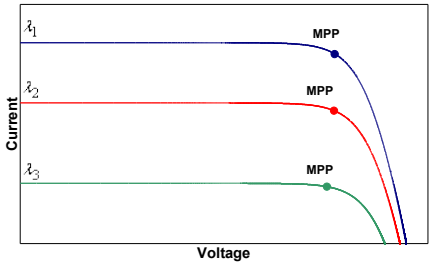
PV Power **plant**

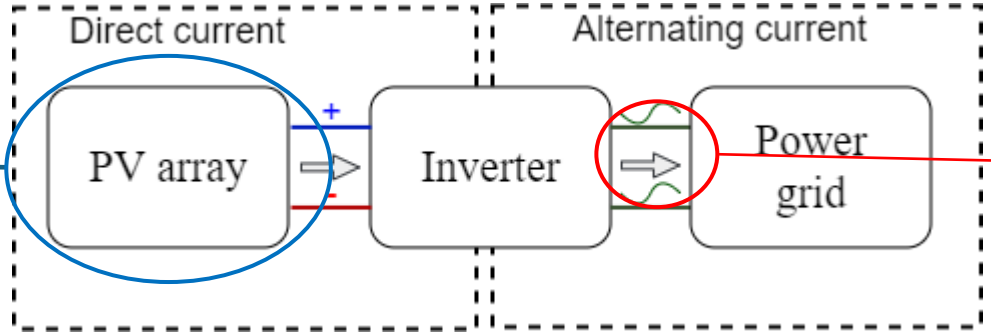
Power grid

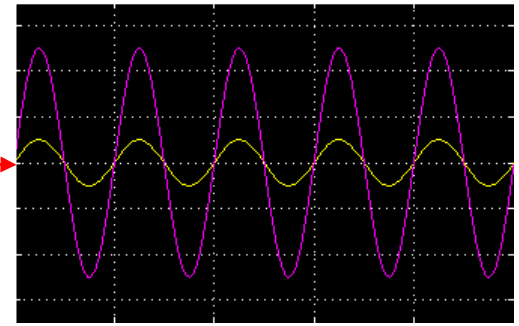
Industry

Commercial

Residential

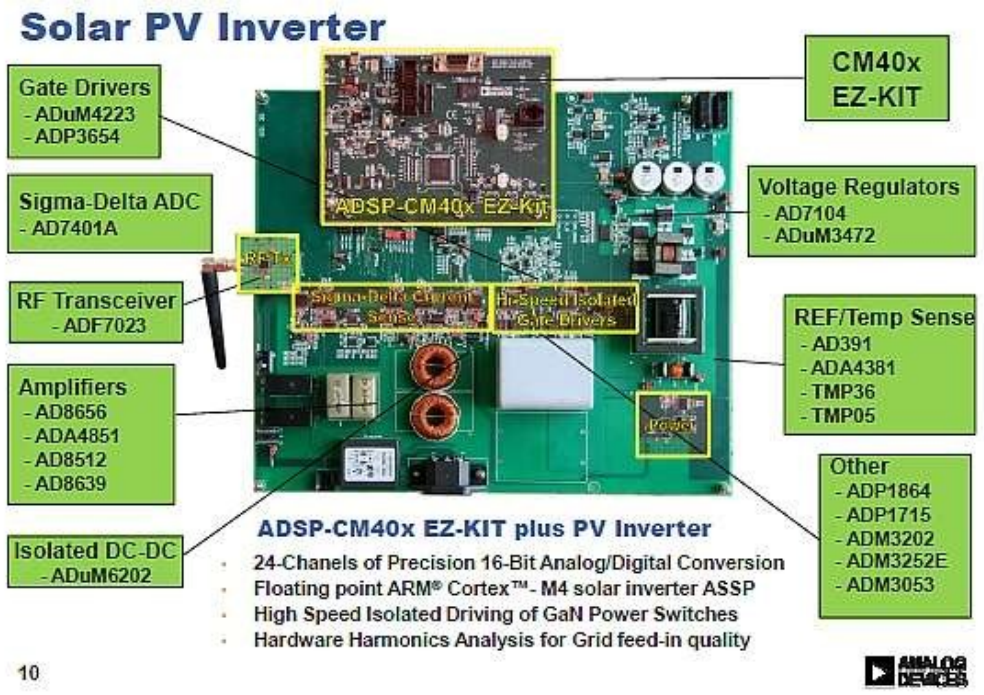


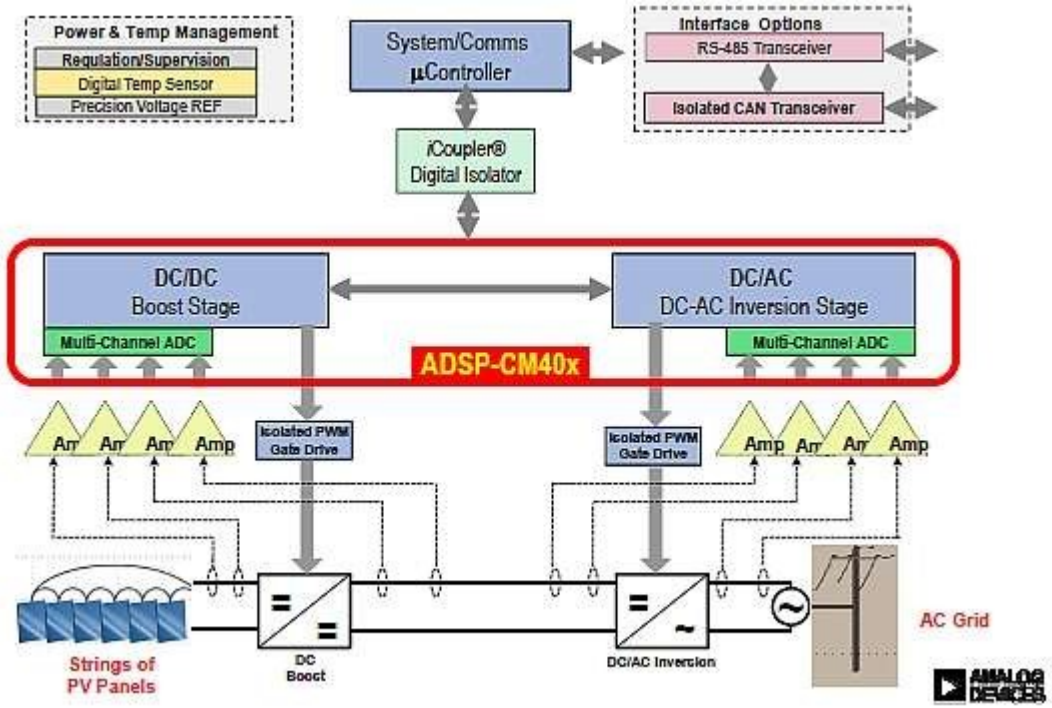




PV Inverter

It is more than a power electronic circuit.





Source: S. Taranovich. Analog Devices mixed-signal control processors for industrial motors and PV inverter designs.

Characteristics of a PV installation

TEC | Tecnológico de Costa Rica

¿Cuáles son las principals características de un Sistema fotovoltaico?

Photovoltaic system characteristics

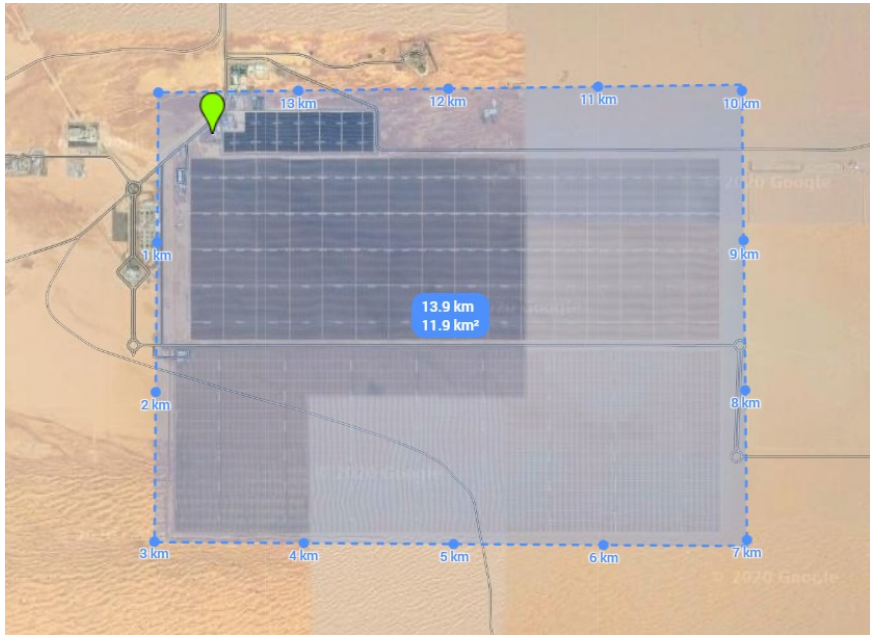
- It has a low energy density, i.e., it requires large surfaces.

- It is modular and can be installed in different ways.

- It comprises several pieces of equipment from different manufacturers with different warranties.

- Power generation depends on the weather, which is a coupled nonlinear chaotic system -> it is not possible to predict in the long term.





Photovoltaic system characteristics

- It has a low energy density, i.e., it requires large surfaces.

- It is modular and can be installed in different ways.

- It comprises several pieces of equipment from different manufacturers with different warranties.

- Power generation depends on the weather, which is a coupled nonlinear chaotic system -> it is not possible to predict in the long term.







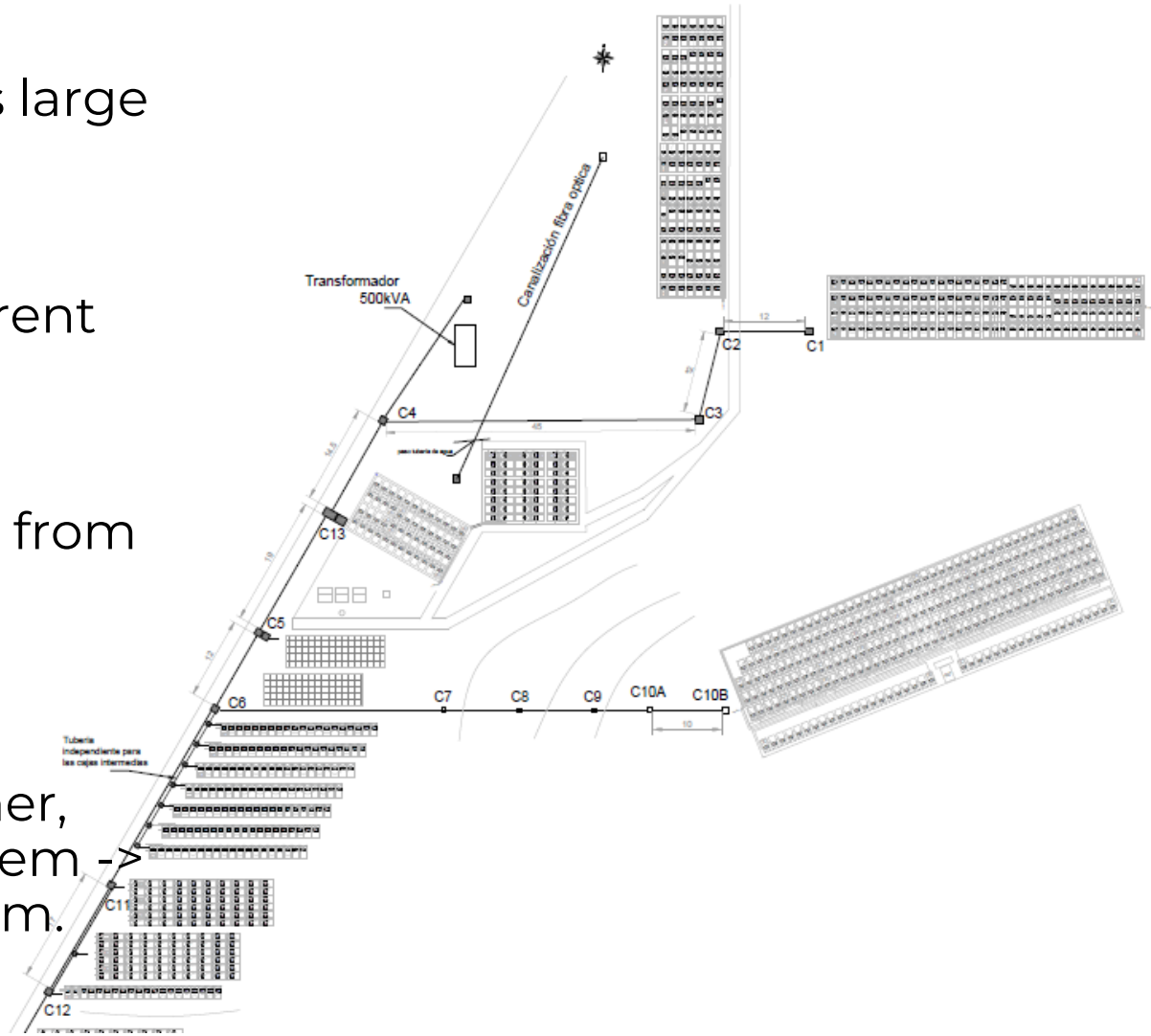
Photovoltaic system characteristics

- It has a low energy density, i.e., it requires large surfaces.

- It is modular and can be installed in different ways.

- It comprises several pieces of equipment from different manufacturers with different warranties.

- Power generation depends on the weather, which is a coupled nonlinear chaotic system -> it is not possible to predict in the long term.



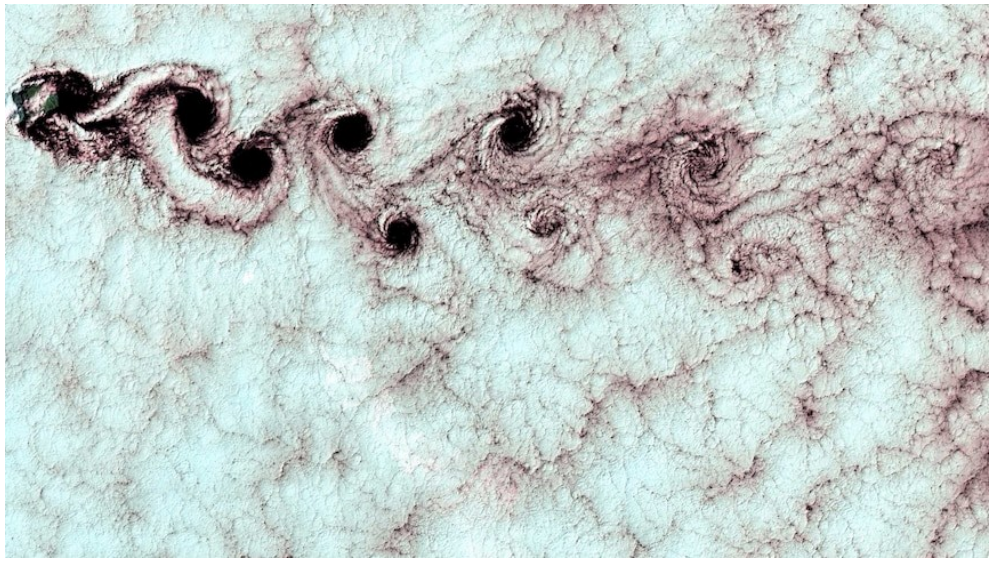
Photovoltaic system characteristics

- It has a low energy density, i.e., it requires large surfaces.

- It is modular and can be installed in different ways.

- It comprises several pieces of equipment from different manufacturers with different warranties.

- Power generation depends on the weather, which is a coupled nonlinear chaotic system -> it is not possible to predict in the long term.



Swirling cloud pattern (Source: USGS EROS Data Center)